

KOBE PRIOR

linkedin.com/in/kobeprior | kobeprior99.github.io | kdprior@mines.edu | (970) 749-9254 | Golden, Colorado

EDUCATION

Colorado School of Mines

May 2027

Master of Science in Electrical Engineering, Focus - Antennas and Communication

GPA: 4.0

- IEEE Eta Kappa Nu Honor Society

Colorado School of Mines

May 2026

Bachelor of Science in Electrical Engineering

GPA: 4.0

- Outstanding Graduating Senior, 6x Dean's List Honors, Mines Undergraduate Research Fellow, VICEROY Scholar

WORK EXPERIENCE

Zeta Associates

May 2026 – August 2026

DSP Intern

Greenwood Village, CO

Utilized four passive collectors to generate a 3D geolocation with a 95% confidence ellipsoid of a sUAS and its controller

- Implemented a FFT-accelerated cross-ambiguity algorithm to determine time difference of arrival between collectors in order to meet 100ms latency requirement with conservative margin
- Leveraged Gauss-Newton optimization to solve nonlinear multilateration systems from CAF-generated time delays
- Designed a graphical user interface to dynamically visualize the geolocation of a sUAS and its controller in real-time

Rincon Research Corporation

May 2025 – July 2025

DSP Intern

Centennial, CO

Developed physics simulation and small scale hardware realization of bent-pipe communication system

- Embedded a CesiumJS-based orbit visualization tool into graphical user interface that rendered satellite trajectories, identifying line-of-sight paths for ideal links between ground stations and the satellites
- Formulated a dynamic channel model informed by satellite position and trajectory data to simulate time delay, path loss, and Doppler shift
- Controlled signal generator and 2 software-defined radios in a small-scale hardware demonstration by issuing asynchronous ZMQ and SSH commands from a graphical user interface

Colorado School of Mines

August 2023 – Present

Resident Advisor

Golden, CO

- Mentored and supported 30+ residents by fostering community, addressing concerns, and collaborating with other RAs

Colorado School of Mines

August 2022 – Present

Student Researcher

Golden, CO

- Characterized scattering due to structured waveforms incident on variable geometry scatterers through full-wave FDTD simulations and physical experimentation

PROJECTS

Low-cost, Software-defined, 16-port, Phased Array

August 2025 – Present

Phased Arrays and Scattering Applications

Colorado School of Mines

- Designed a custom 16-port digital phase-shifting network in Altium Designer, creating hierarchical schematics, custom footprints, and schematic symbols for power dividers/combiners and 8-bit phase shifters
- Developed the complete control architecture, including STM32 firmware for serial communication and addressing of phase shifters and a Python GUI for manual phase control and automated beam-scanning configurations
- Demonstrated brute force direction-finding and generation of structured waveforms, presenting results at the VICEROY Symposium and the 2026 National Radio Science Meeting

Low-Noise Amplifier

January 2026 – May 2026

Active RF and Microwave Devices

Colorado School of Mines

- Designed and fabricated an LNA in Keysight ADS achieving >10 dB gain and <3 dB noise figure at 2.5 GHz
- Characterized the amplifier with a vector network analyzer and spectrum analyzer to verify gain and linearity

SKILLS

Languages: Python, C++, Embedded C, MATLAB, Verilog, Arduino, HTML, LaTeX

Software: Keysight ADS, XMidass, Altium Designer, HFSS, Vim, Git, Linux, GNU Radio, Microsoft Tools

Testing Equipment: Vector Network Analyzer, Oscilloscope, Digital Multimeter, Spectrum Analyzer, Function Generator

Hardware: Arduino, STM32, SDR, LPKF, FPGA, 3D Printer, Printed Circuit Board Design